Results

Every number below is copied from results_tables.md / `results/summary.json`, which `scripts/collect_results.py` regenerates from the run directories under `runs/` (protocol: EXPERIMENTS.md; data: DATA.md). Generation metrics use the `best/` weights (lowest validation diffusion loss, EMA) and 5,000 DDIM-50 samples unless stated; segmentation Dice is per-patient on the 74 held-out test patients, mean ± std over 3 seeds.

Status (2026-09-10 23:00): the diffusion models, the checkpoint curves, the model selection, the 128 px segmentation study with its secondary analyses and the pre-training control, and the 256 px segmentation study are complete. The pre-declared VAE decoder fine-tune study (section 7) is running: the decoder is trained and its ceiling measured (7.1), the re-decoded samples are scored (7.2) and the primary segmentation protocol is repeated (7.3); its secondary analyses (27 U-Nets, ≈ 3 h) are marked *pending*.

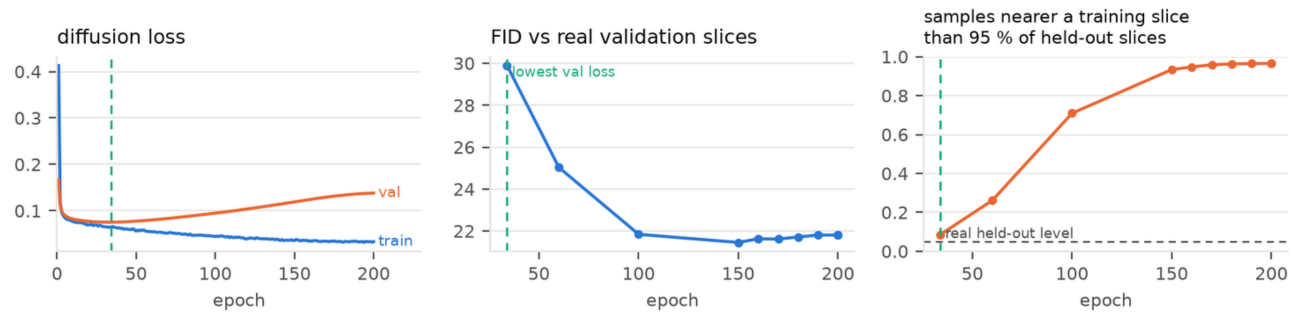
Headline findings

1. **Without regularisation the latent diffusion model memorises.** LDM-128-mask reaches its lowest validation loss at epoch 34; by epoch 200 its FID against real validation slices has improved from 29.9 to 21.8, but 97 % of its samples then lie closer to a training slice (either orientation) than 95 % of real held-out slices do. The "better" late checkpoints are copies.
2. **Cached affine augmentation, not dropout, fixes this.** With dropout 0.1 + 6 augmented latent variants per slice (LDM-128-mask-reg) the validation loss keeps falling to epoch 101, the memorised fraction at the last epoch is 0.115 instead of 0.966, and the early-stopped model is the best of the three candidates on validation data (FID 24.7 vs 28.8 / 29.9). Dropout alone changes nothing (0.891 memorised at epoch 200). The selection was made on validation slices only.
3. **At 256 px the regularised model is near the resolution's floor.** LDM-256-mask-reg: FID 10.45 vs the real test slices, where two disjoint sets of *real* slices (val vs test) score 9.55; memorised 0.030; 10.83 with tumour masks of patients it never saw.
4. **Mixing synthetic slices into training did not improve the 128 px tumour segmenter.** Adding patient-matched synthetic pairs raises whole-tumour Dice slightly at 10 % real data (0.801 → 0.809) but lowers tumour-core and enhancing-tumour Dice at every data fraction (ET at 10 %: 0.509 → 0.430; mean Dice at 100 %: 0.778 → 0.757). A segmenter trained on synthetic data alone reaches 0.660 mean Dice, about the level of 10 % real data.
5. **The loss is the frozen VAE's, not the diffusion model's.** Real slices merely passed through the frozen VAE (no generator) cost more Dice than the synthetic slices do (−0.057 to −0.072 mean, ET −0.11 to −0.14), and halving the synthetic ratio barely helps. The natural-image decoder removes enhancing-tumour detail; every synthetic image inherits that.
6. **Used as pre-training instead of as extra training data, the synthetic pairs help when real data is scarce, even against a compute-matched control**: +0.021 mean Dice at 10 % real and +0.014 at 25 % over a segmenter given the same extra epochs on real data (raw gains over real only: +0.044 and +0.015); nothing at 100 %.
7. **At 256 px, where the generator sits at the FID floor and the VAE ceiling is 2.6 dB higher, the mixed synthetic data helps in the low-data regime**: +0.027 mean Dice with 26 real patients (all regions, 3/3 seeds), neutral with 64 and with all 258. Synthetic-only training reaches the level of 25 % real data.

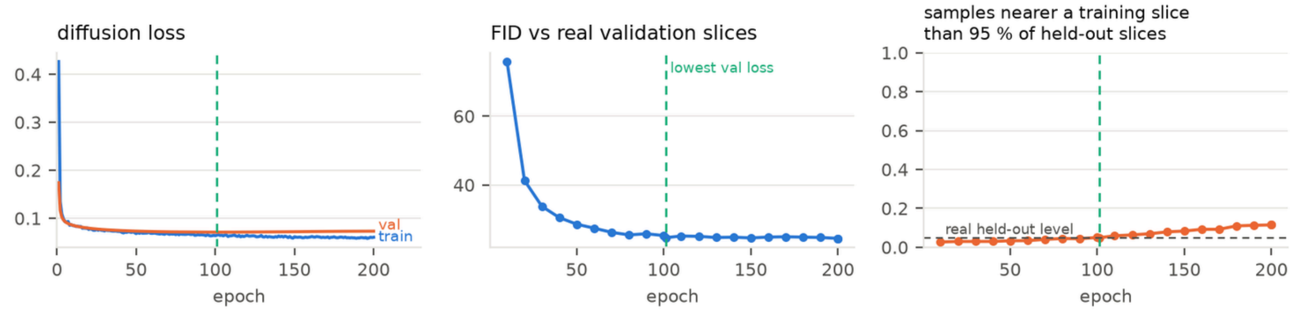
1. Memorisation vs training length (checkpoint curves)

2,000 samples per saved checkpoint, fixed masks and seeds; FID against the real *validation* slices; "memorised" = fraction of samples closer to a training slice (both orientations) than 95 % of real held-out slices are. Figures: `figures/<run>_checkpoint_curve.png` (loss / FID / memorised fraction per epoch).

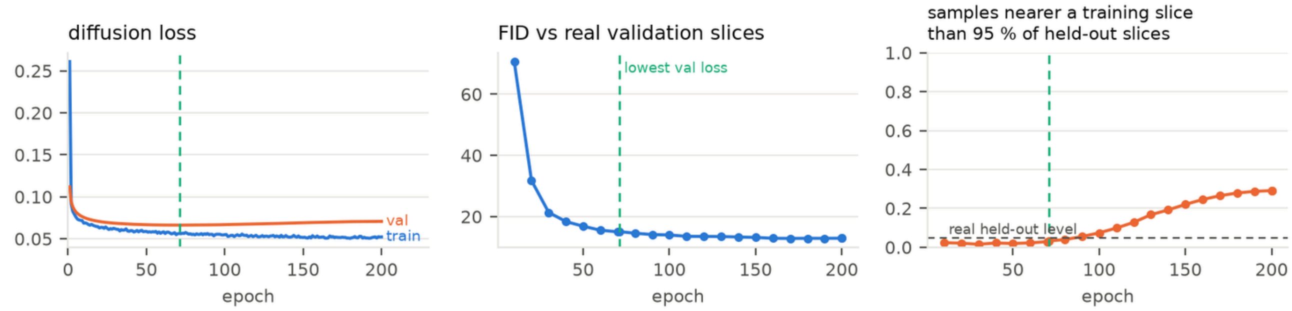
ldm128_maskcond



ldm128_maskcond_reg



ldm256_maskcond_reg



run	best epoch (val loss)	FID val at best	memorised at best	FID val at epoch 200	memorised at epoch 200
LDM-128-mask (baseline)	34	29.89	0.086	21.81	**0.966**
LDM-128-mask-do0.1	41	28.82	0.093	21.99	**0.891**
LDM-128-mask-reg	101	24.65	0.049	24.43	0.115
LDM-128-reg (unconditional)	116	24.81	0.040	23.20	0.051
LDM-256-mask-reg	71	15.10	0.033	12.94	0.291

The FID-vs-epoch curve alone would favour the last checkpoint of every run; the nearest-neighbour check shows that the FID gain after the validation-loss minimum is bought with copies. Reporting from the best-validation-loss checkpoint is therefore the protocol for everything below. Even the regularised 256 px model starts copying after epoch ≈ 90 (0.074 at epoch 100, 0.291 at epoch 200), so early stopping stays necessary; augmentation postpones memorisation, it does not remove it.

2. Regularisation ablation and model selection (validation data only)

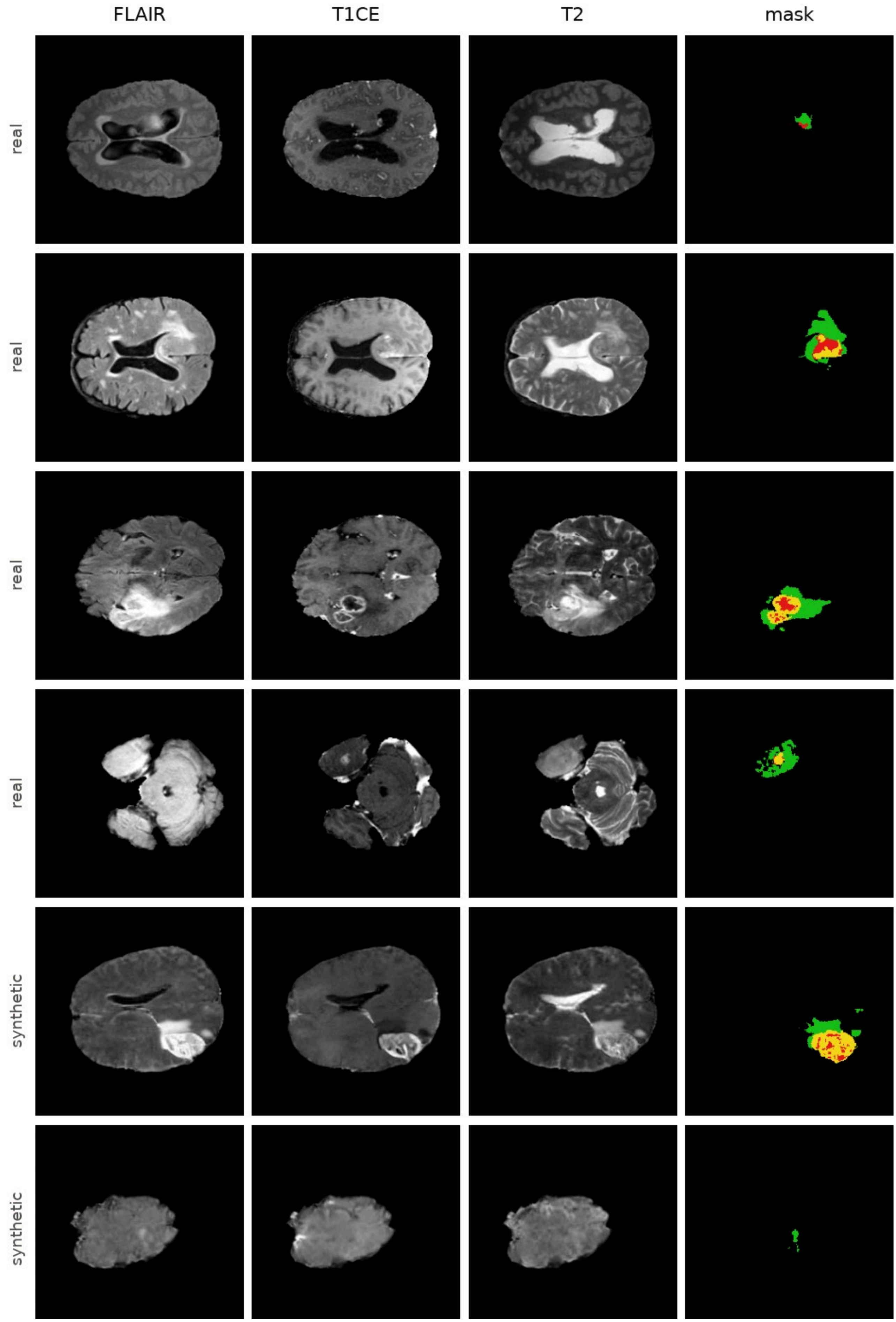
Rule fixed before the runs finished: lowest FID against validation slices at the best-val-loss checkpoint, among candidates with memorised ≤ 0.15 (`results/model_selection.json`).

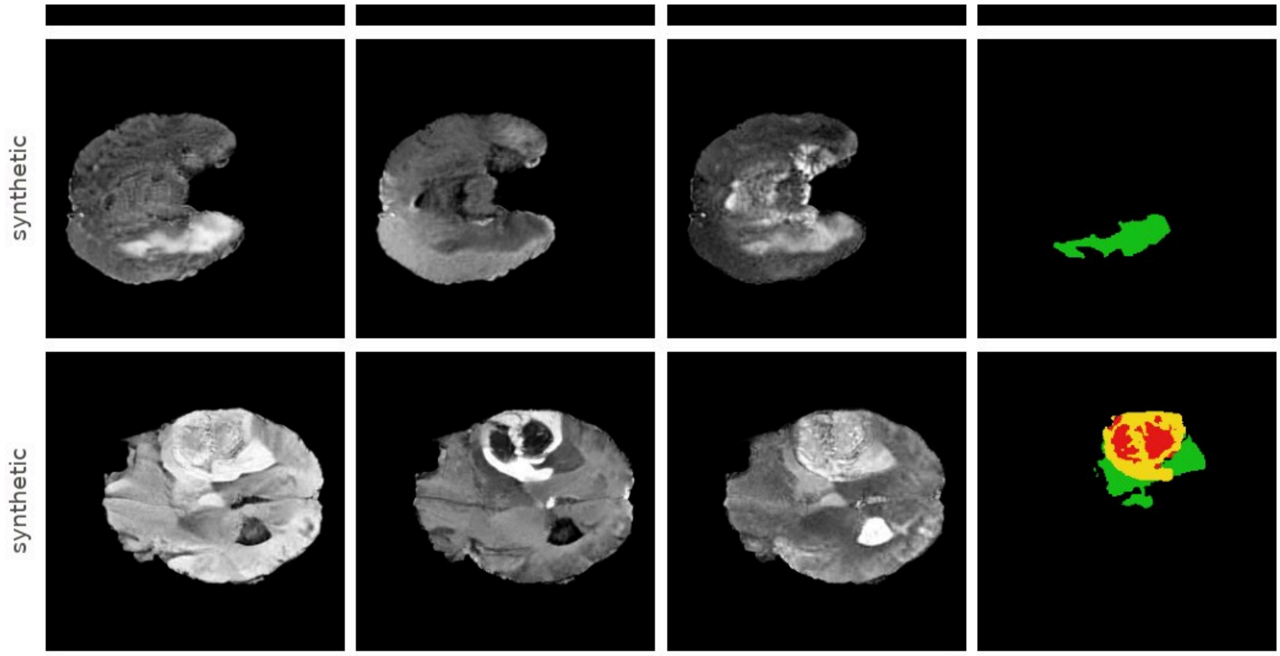
candidate	best epoch	FID val ↓	KID val ×10³ ↓	memorised ↓
baseline (flips only)	34	29.89	18.18	0.086
+ dropout 0.1	41	28.82	17.24	0.093
+ dropout 0.1 + affine augmentation **(chosen)**	101	**24.65**	**12.86**	**0.049**

The chosen recipe was then reused unchanged for the unconditional (LDM-128-reg) and the 256 px (LDM-256-mask-reg) models, and its 20,000-sample pool feeds the segmentation study.

3. Generation quality on the test set

Against all 4,623 real test slices. `cfg` = classifier-free guidance scale; `val masks` = samples conditioned on masks of validation patients (tumour shapes never seen in training). Reference floor: FID between real validation and real test slices = 10.49 (128 px) / 9.55 (256 px). Pair-SSIM is the mean SSIM of 2,000 random sample pairs (lower = more diverse; real test slices 0.652 / 0.690). Side-by-side sheets: `figures/<run>_<samples>_real_vs_synth.png`.


FLAIR
T1CE
T2
mask
real
real
real
real
synthetic
synthetic



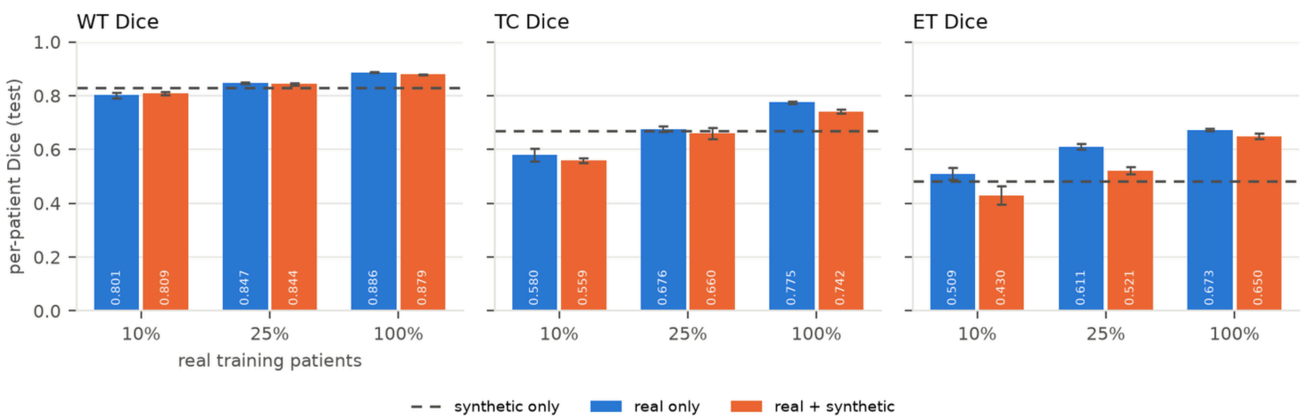
model	setting	FID rgb ↓	KID ×10³ ↓	FID flair / t1ce / t2	pair-SSIM	memorised ↓
LDM-128-mask (baseline)	cfg 2	26.22	20.89	45.9 / 45.4 / 73.0	0.576	0.069
LDM-128-mask-do0.1	cfg 2	24.34	19.13	47.7 / 46.2 / 75.7	0.585	0.070
LDM-128-mask-reg	cfg 2	20.85	15.19	44.3 / 44.0 / 69.6	0.584	0.043
LDM-128-mask-reg	cfg 1	20.37	14.74	47.0 / 44.2 / 69.4	0.586	0.036
LDM-128-mask-reg	cfg 2, val masks	22.85	16.52	45.2 / 44.1 / 71.1	0.593	0.040
LDM-128-reg (unconditional)	cfg 1	21.31	15.87	47.5 / 45.0 / 68.7	0.579	0.027
LDM-256-mask-reg	cfg 2	**10.45**	**6.40**	40.3 / 39.6 / 59.1	0.628	0.030
LDM-256-mask-reg	cfg 1	12.83	8.77	43.1 / 41.7 / 58.2	0.637	0.021
LDM-256-mask-reg	cfg 2, val masks	10.83	6.17	40.4 / 39.3 / 60.7	0.633	0.024

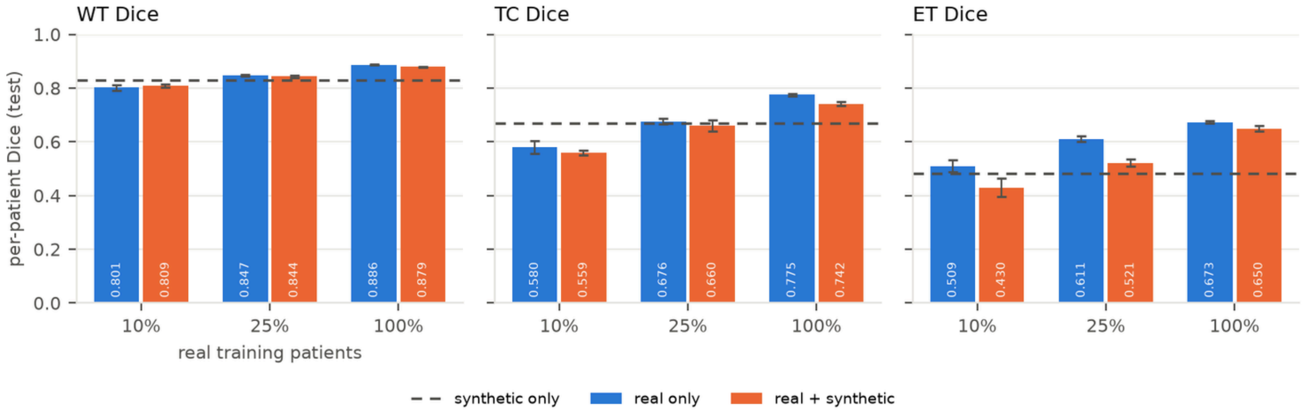
Observations: (i) the regularised recipe improves test FID by 5.4 points over the baseline at the same resolution; (ii) unseen validation masks cost 2.0 FID points at 128 px and 0.4 at 256 px, so the models generalise to new tumour geometry; (iii) guidance 2 helps at 256 px (10.45 vs 12.83) and is slightly worse than guidance 1 at 128 px (20.85 vs 20.37); (iv) per-modality FIDs are far above the composite because each grey channel is replicated to RGB for Inception, and T2 is consistently the hardest modality; (v) all memorised fractions sit at or below the 0.05 expected for a model that generalises; (vi) the mask-conditioned and unconditional 128 px models score alike, so conditioning costs no fidelity.

VAE ceiling (`decode(encode(x))` on real test slices, frozen `sd-vae-ft-mse`): PSNR 26.4 dB / SSIM 0.833 / LPIPS 0.041 at 128 px and 29.0 dB / 0.877 / 0.036 at 256 px (T2 is the worst channel at both). No latent model can be sharper than this; part of the fine-detail loss discussed below is the autoencoder's, not the diffusion model's.

4. Downstream segmentation at 128 px (primary protocol)

2-D U-Net, 40 epochs, model selection on real validation patients, per-patient Dice on the 74 test patients, 3 seeds. Synthetic slices come from LDM-128-mask-reg (`best/`, cfg 2) and are patient-matched: a real-x % segmenter only receives synthetic slices conditioned on masks of its own x % patients. Figure: `figures/segmentation_dice.png`.





training data	real patients	real slices	synthetic slices	WT	TC	ET	mean
10 % real	26	1,483–1,705	0	0.801 ± 0.011	0.580 ± 0.023	0.509 ± 0.022	0.630 ± 0.016
10 % real + synthetic	26	1,483–1,705	1,845–2,208	0.809 ± 0.006	0.559 ± 0.010	0.430 ± 0.033	0.600 ± 0.014
25 % real	64	3,937–4,031	0	0.847 ± 0.003	0.676 ± 0.011	0.611 ± 0.010	0.711 ± 0.004
25 % real + synthetic	64	3,937–4,031	4,913–5,158	0.844 ± 0.004	0.660 ± 0.021	0.521 ± 0.014	0.675 ± 0.012
100 % real	258	15,895	0	0.886 ± 0.001	0.775 ± 0.005	0.673 ± 0.004	0.778 ± 0.003
100 % real + synthetic	258	15,895	20,000	0.879 ± 0.001	0.742 ± 0.009	0.650 ± 0.010	0.757 ± 0.006
synthetic only	0	0	20,000	0.829 ± 0.002	0.669 ± 0.007	0.481 ± 0.012	0.660 ± 0.005

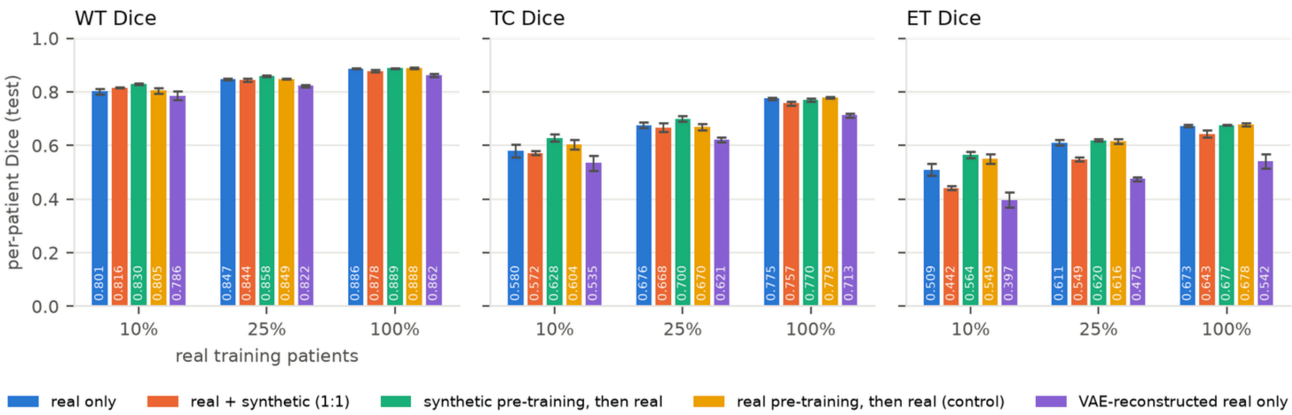
Mask consistency (per-slice Dice between a real-trained segmenter's prediction on a synthetic image and the mask it was conditioned on, all 20,000 pairs):

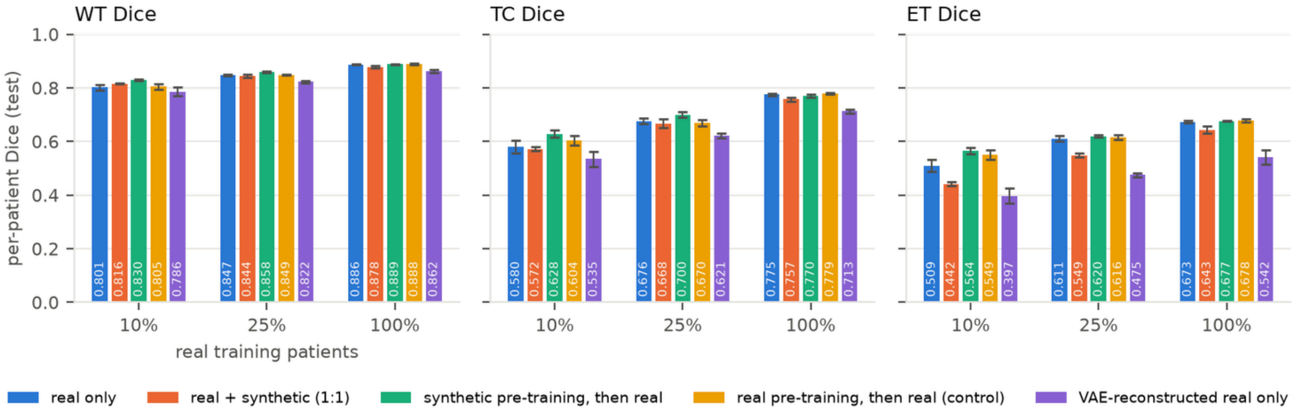
segmenter trained on	WT	TC	ET
10 % real	0.767 ± 0.006	0.506 ± 0.023	0.439 ± 0.025
25 % real	0.783 ± 0.003	0.566 ± 0.006	0.493 ± 0.012
100 % real	0.789 ± 0.002	0.584 ± 0.006	0.494 ± 0.004

Reading: the synthetic pairs carry the whole-tumour outline well (WT consistency 0.79 vs a real test Dice of 0.89 for the same segmenter) but the enhancing-tumour appearance inside the mask matches the label only about half the time. Mixing such pairs into training teaches the segmenter a blurred ET/TC appearance, which is why the loss is largest on ET and shows up even with 100 % real data. Synthetic-only training reaching 0.660 confirms that the pairs are label-consistent enough to be usable, but not sharper than what 26 real patients provide.

5. Secondary segmentation analyses (128 px)

Declared in EXPERIMENTS.md after the first seed-0 results, run with the same 3 seeds, patient matching and 40-epoch schedule (`runs/seg/real<pct>_{synth1x,vae,pre}_s<seed>`). Figure: `figures/segmentation_dice_secondary.png`.





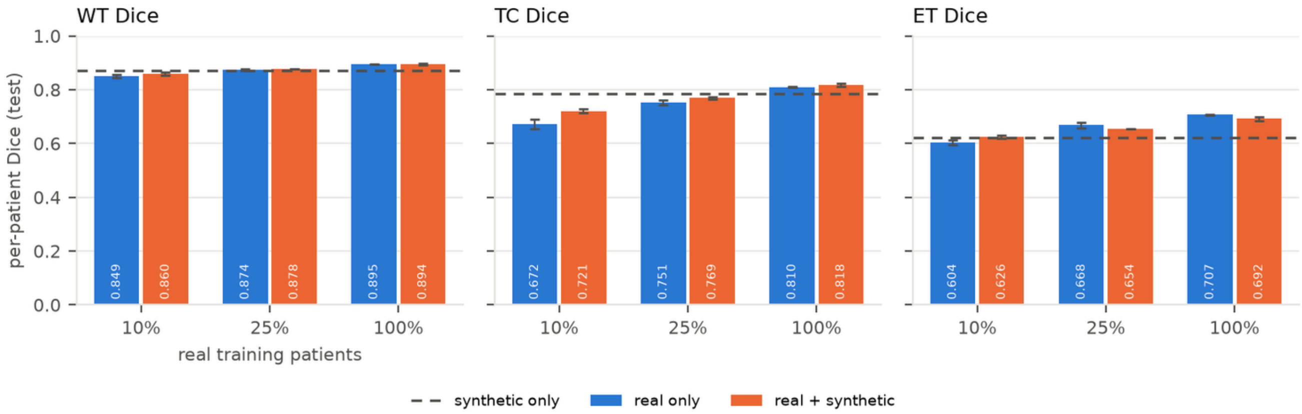
training data	WT	TC	ET	mean	Δ mean vs real only
10 % real	0.801 ± 0.011	0.580 ± 0.023	0.509 ± 0.022	0.630 ± 0.016	
10 % real + synthetic (1.25:1, primary)	0.809 ± 0.006	0.559 ± 0.010	0.430 ± 0.033	0.600 ± 0.014	−0.030
10 % real + synthetic 1:1	0.816 ± 0.002	0.572 ± 0.008	0.442 ± 0.007	0.610 ± 0.006	−0.020
10 % real, VAE-reconstructed	0.786 ± 0.016	0.535 ± 0.029	0.397 ± 0.028	0.573 ± 0.023	−0.057
10 % real, synthetic pre-training	0.830 ± 0.003	0.628 ± 0.013	0.564 ± 0.012	0.674 ± 0.008	**+0.044**
25 % real	0.847 ± 0.003	0.676 ± 0.011	0.611 ± 0.010	0.711 ± 0.004	
25 % real + synthetic (1.25:1, primary)	0.844 ± 0.004	0.660 ± 0.021	0.521 ± 0.014	0.675 ± 0.012	−0.036
25 % real + synthetic 1:1	0.844 ± 0.007	0.668 ± 0.018	0.549 ± 0.008	0.687 ± 0.011	−0.024
25 % real, VAE-reconstructed	0.822 ± 0.005	0.621 ± 0.009	0.475 ± 0.006	0.639 ± 0.003	−0.072
25 % real, synthetic pre-training	0.858 ± 0.003	0.700 ± 0.010	0.620 ± 0.006	0.726 ± 0.005	**+0.015**
100 % real	0.886 ± 0.001	0.775 ± 0.005	0.673 ± 0.004	0.778 ± 0.003	
100 % real + synthetic (1.25:1, primary)	0.879 ± 0.001	0.742 ± 0.009	0.650 ± 0.010	0.757 ± 0.006	−0.021
100 % real + synthetic 1:1	0.878 ± 0.004	0.757 ± 0.007	0.643 ± 0.013	0.760 ± 0.004	−0.018
100 % real, VAE-reconstructed	0.862 ± 0.006	0.713 ± 0.008	0.542 ± 0.026	0.706 ± 0.006	−0.072
100 % real, synthetic pre-training	0.889 ± 0.002	0.770 ± 0.007	0.677 ± 0.001	0.779 ± 0.003	+0.001

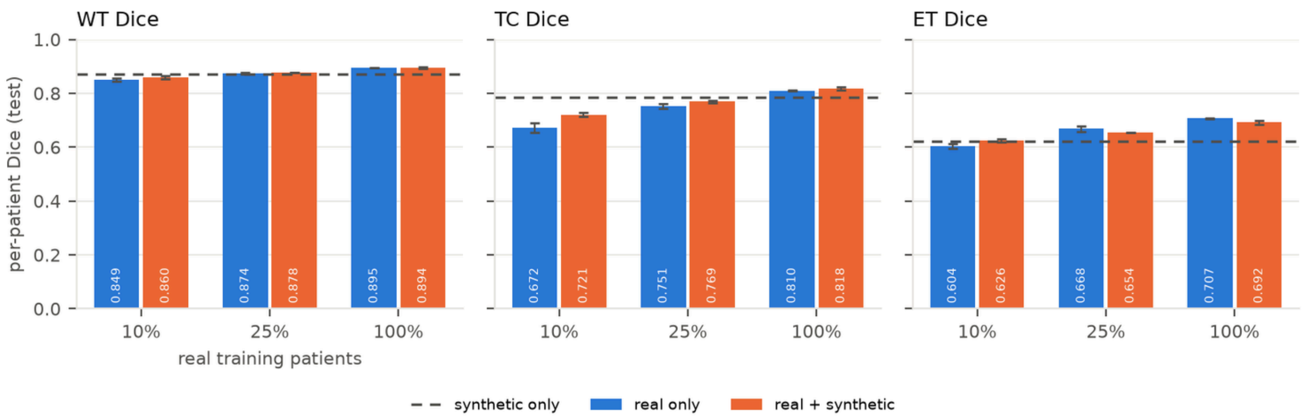
Three answers to the questions posed in the declaration:

- **The synthetic ratio is not the cause.** Capping synthetic slices at 1:1 recovers only about a third of the loss (mean −0.020 / −0.024 / −0.018 instead of −0.030 / −0.036 / −0.021); ET stays 0.03–0.07 below real only at every fraction.
- **The frozen VAE alone reproduces the loss, and more.** Real slices passed through `decode(encode(x))` of the frozen `sd-vae-ft-mse`, with no generator involved, cost −0.057 / −0.072 / −0.072 mean Dice, again concentrated on ET (−0.11 to −0.14). Every synthetic image carries exactly this decoder, so the enhancing-tumour detail the segmenter needs is removed before the diffusion model is even involved. This motivates the decoder fine-tune of section 7.
- **Pre-training on synthetic pairs, then fine-tuning on real, helps in the low-data regime.** +0.044 mean Dice at 10 % real (all three regions, ET 0.509 → 0.564, 3/3 seeds), +0.015 at 25 %, nothing at 100 %. Fine-tuning on real slices overrides the blurred appearance that mixing bakes in, while the pre-trained features still transfer. The pre-trained runs receive 20 extra epochs; section 8 shows that a control given the same extra epochs on real data recovers about half of the 10 % gain and none of the 25 % gain, so the synthetic-specific effect is +0.021 / +0.014.

6. Segmentation study at 256 px

Same protocol as section 4 (declared before any 256 px segmenter was trained) with the 256 px slices and the LDM-256-mask-reg pool (`runs/seg256/`, 21 U-Nets). Figure: `figures/segmentation_dice_seg256.png`.





training data	WT	TC	ET	mean	Δ mean vs real only
10 % real	0.849 ± 0.006	0.672 ± 0.019	0.604 ± 0.009	0.709 ± 0.003	
10 % real + synthetic	0.860 ± 0.006	0.721 ± 0.007	0.626 ± 0.006	0.736 ± 0.004	**+0.027**
25 % real	0.874 ± 0.003	0.751 ± 0.009	0.668 ± 0.010	0.764 ± 0.007	
25 % real + synthetic	0.878 ± 0.000	0.769 ± 0.004	0.654 ± 0.000	0.767 ± 0.001	+0.003
100 % real	0.895 ± 0.000	0.810 ± 0.002	0.707 ± 0.002	0.804 ± 0.000	
100 % real + synthetic	0.894 ± 0.004	0.818 ± 0.006	0.692 ± 0.008	0.801 ± 0.005	−0.003
synthetic only	0.871 ± 0.002	0.785 ± 0.008	0.623 ± 0.004	0.760 ± 0.003	

At 256 px the sign flips in the low-data regime: with 26 real patients, adding the patient-matched synthetic pairs improves every region (TC +0.049, ET +0.022, WT +0.011; 3/3 seeds, spread ≤ 0.004), where the same protocol at 128 px lost 0.030. With 64 patients the effect is neutral (TC up, ET down), and with all 258 it is neutral on the mean and still slightly negative on ET (−0.015). The synthetic-only segmenter reaches 0.760, the level of 25 % real data (128 px: 0.660, the level of 10 %). The 256 px pool comes from a generator at the FID floor (10.45 vs 9.55) decoded through a VAE whose ceiling is 29.0 dB / SSIM 0.877 instead of 26.4 dB / 0.833, consistent with section 5's finding that the decoder's loss of detail, not the diffusion model, was limiting the 128 px result. Note that the 256 px real-only segmenters are themselves stronger (0.709 / 0.764 / 0.804 vs 0.630 / 0.711 / 0.778), so the two resolutions are compared only within themselves.

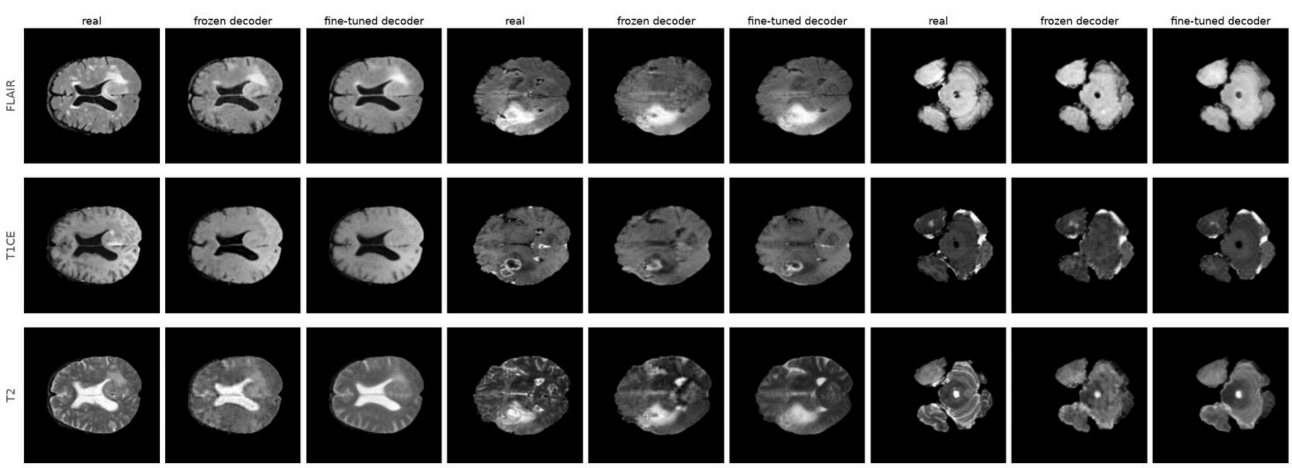
7. VAE decoder fine-tuning (running: 7.1–7.2 and the primary part of 7.3 done, secondary *pending*)

Declared after section 5's VAE control (protocol and pre-declared readings in EXPERIMENTS.md). Only `decoder` + `post_quant_conv` of `sd-vae-ft-mse` (49.5 M parameters) were trained on the 15,895 training slices with L1 + 0.5·LPIPS-VGG per channel (AdamW 2e-5, batch 16, horizontal flips, 8 epochs, 57 min on the A6000); the epoch with the lowest validation loss is kept (epoch 8, 0.096 → 0.067, still falling slowly). The encoder is untouched, so every cached latent and LDM-128-mask-reg itself stay valid: the same latents are simply decoded again (`runs/vae_dec_brats128/`, `scripts/finetune_vae_decoder.py`).

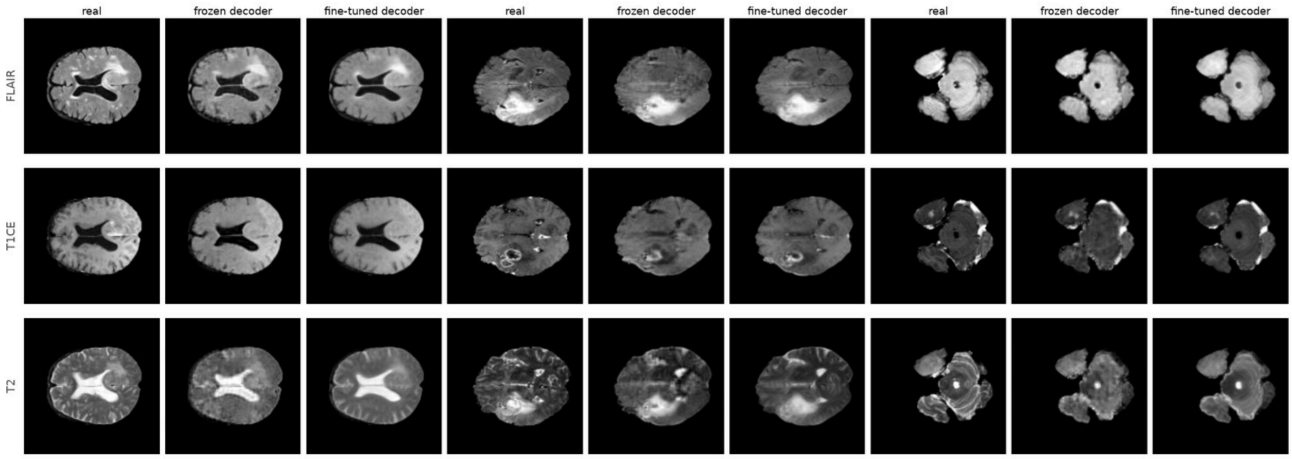
7.1 Reconstruction ceiling on the 4,623 test slices (pre-declared reading a)

decoder	PSNR ↑	SSIM ↑	LPIPS-Alex ↓	FLAIR / T1ce / T2 PSNR
frozen `sd-vae-ft-mse`	26.37 dB	0.833	0.041	27.0 / 27.2 / 25.4
fine-tuned decoder	**27.79 dB**	**0.883**	**0.032**	28.6 / 28.3 / 27.0

All three metrics improve, so reading (a) is passed and the study continues. The gain is about half of the gap to the 256 px ceiling in PSNR (29.0 dB) and exceeds it in SSIM (0.877); T2, the worst channel of the frozen decoder, gains the most (+1.6 dB, SSIM 0.817 → 0.880). The figure `figures/vae_decoder_brats128.png` shows three test slices with enhancing tumour through both



decoders: the blotchy texture the frozen decoder puts into T2 disappears and the enhancing rims in T1ce come out sharper.

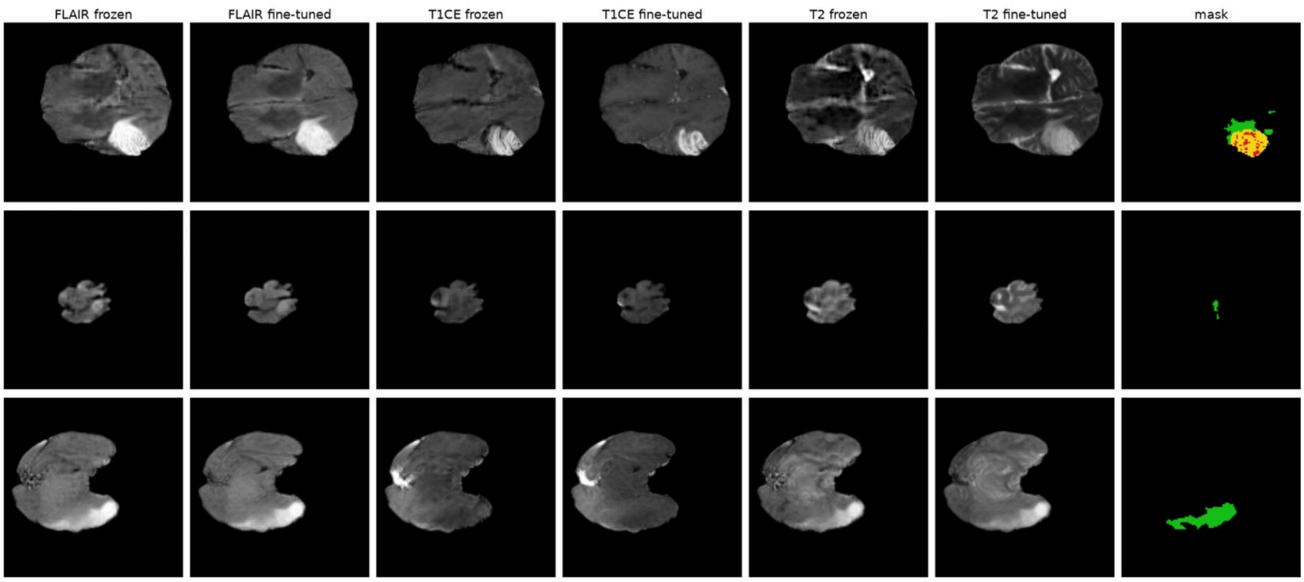


7.2 The same samples decoded with the fine-tuned decoder

The 20,000 training-mask, 5,000 g = 1 and 5,000 validation-mask sample sets of LDM-128-mask-reg were decoded again from the *same* latents (`samples_*_ftdec`) and scored as in section 3 (5,000 samples per score). FID/KID rgb score the FLAIR/T1ce/T2 stack as one colour image; the per-modality FIDs score each grey channel on its own.

sample set	decoder	FID rgb ↓	KID rgb $\times 10^3$ ↓	FID FLAIR / T1ce / T2 ↓	pair-SSIM (real pairs 0.652)	NN dist to train (real test 9.96)	memorised ↓
training masks, g = 2	frozen	20.85	15.19	44.3 / 44.0 / 69.6	0.584	10.71	0.043
	fine-tuned	20.89	18.91	**28.6 / 25.6 / 17.6**	0.636	10.49	0.051
training masks, g = 1	frozen	20.37	14.74	47.0 / 44.2 / 69.4	0.586	10.60	0.036
	fine-tuned	18.06	14.56	**26.8 / 25.7 / 17.5**	0.638	10.46	0.040
validation masks, g = 2	frozen	22.85	16.52	45.2 / 44.1 / 71.1	0.593	11.09	0.040
	fine-tuned	22.49	20.21	**29.3 / 26.7 / 17.9**	0.640	10.90	0.046

The composite FID hardly moves (20.85 → 20.89, 20.37 → 18.06, 22.85 → 22.49) and KID rgb gets slightly worse for two sets, but the per-modality FIDs fall by 35–75 %: T2 69.6 → 17.6, T1ce 44.0 → 25.6, FLAIR 44.3 → 28.6. The frozen decoder's per-channel scores were dominated by the texture it invents (compare the T2 columns in the figure below), which the composite metric, seeing the three channels as one colour image, largely ignores; the per-modality numbers are the ones a radiologist would recognise. Sample diversity moves towards that of real slices (pairwise SSIM 0.584 → 0.636 vs 0.652 for real pairs), so part of the frozen decoding's apparent diversity was decoder noise. The samples also come slightly closer to the training set (mean nearest-neighbour distance 10.71 → 10.49; real held-out slices 9.96) and the memorised fraction rises from 0.043 to 0.051 with the threshold unchanged. The latents are identical, so this is the sharper decoding moving every sample closer to every real slice, not new copying; the fraction stays at the 0.05 that real held-out slices score by construction.



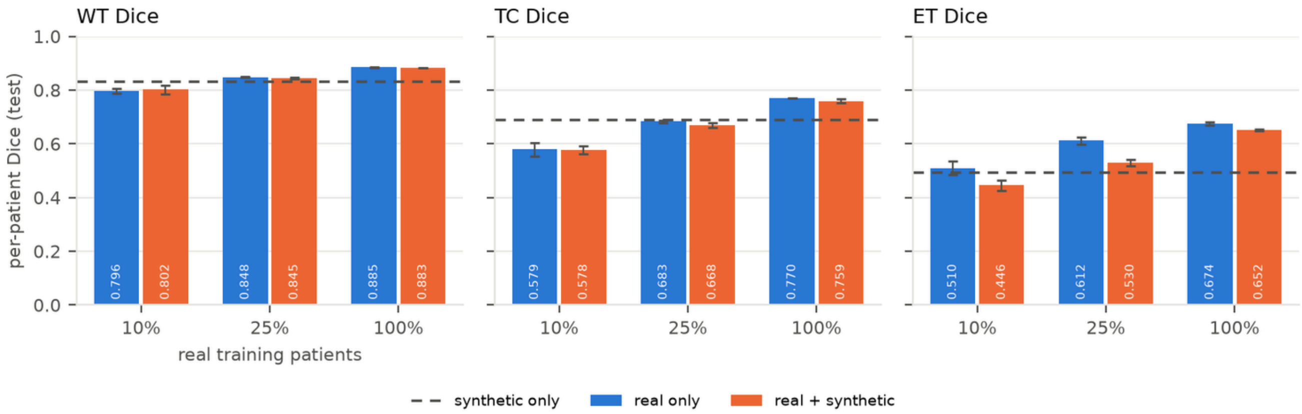
7.3 Segmentation with the re-decoded pool (pre-declared reading b)

The primary protocol of section 4 was repeated unchanged with the re-decoded pool (`runs/seg_ftdec`, same seeds and patient-matched pairs). The real-only segmenters were re-trained too and reproduce section 4 within the seed spread (0.628 / 0.714 / 0.776 vs 0.630 /

0.711 / 0.778 mean Dice), which calibrates the run-to-run noise of everything below.

real patients	decoder	real only	+ synthetic	Δ mean (seed-matched)	Δ WT / TC / ET
26 (10 %)	frozen	0.630 ± 0.016	0.600 ± 0.014	−0.031 (−0.033, −0.020, −0.039)	+0.008 / −0.020 / −0.079
	fine-tuned	0.628 ± 0.017	0.609 ± 0.006	**−0.020** (−0.036, −0.005, −0.019)	+0.006 / −0.001 / −0.064
64 (25 %)	frozen	0.711 ± 0.004	0.675 ± 0.012	−0.036 (−0.046, −0.027, −0.037)	−0.004 / −0.016 / −0.089
	fine-tuned	0.714 ± 0.005	0.681 ± 0.007	**−0.033** (−0.040, −0.031, −0.028)	−0.003 / −0.014 / −0.081
258 (100 %)	frozen	0.778 ± 0.003	0.757 ± 0.006	−0.021 (−0.021, −0.032, −0.011)	−0.007 / −0.033 / −0.024
	fine-tuned	0.776 ± 0.002	0.765 ± 0.004	**−0.012** (−0.018, −0.009, −0.007)	−0.002 / −0.011 / −0.022
synthetic only	frozen	–	0.660 ± 0.005		
	fine-tuned	–	0.672 ± 0.004		

The sharper decoding removes about a third of the loss at 10 % and 100 % real (−0.031 → −0.020, −0.021 → −0.012) and almost none at 25 % (−0.036 → −0.033). The loss stays negative in 9 of 9 seed-matched pairs and enhancing tumour still loses 0.06–0.08 Dice; tumour core is what recovers (10 %: −0.020 → −0.001; 100 %: −0.033 → −0.011). A segmenter trained on the re-decoded synthetic slices alone reaches 0.672 mean Dice (0.660 with the frozen decoding), still the level of 10 % real data. So the decoder's blur was part of the problem but not most of it. The encoder is unchanged, so whether the remaining loss is the latent bottleneck's (detail the 16 × 16 × 4 latent never held, which no decoder can restore) or the diffusion model's is what the secondary study's VAE-reconstructed-real control with the fine-tuned decoder decides (*pending*, 27 U-Nets, ≈ 01:45).



8. Compute-matched control for synthetic pre-training

The synthetic-pre-training runs of section 5 train for 20 + 40 epochs. `runs/seg/real<pct>_prereal_s<seed>` pre-trains for the same 20 epochs on the real slices themselves and then fine-tunes identically (same seeds and fractions), so the two differ only in *what* the 20 extra epochs see.

real patients	real only	+ real pre-training (control)	+ synthetic pre-training	synthetic − control
26 (10 %)	0.630 ± 0.016	0.653 ± 0.010	0.674 ± 0.008	**+0.021** (WT +0.025, TC +0.024, ET +0.015)
64 (25 %)	0.711 ± 0.004	0.712 ± 0.003	0.726 ± 0.005	**+0.014** (WT +0.009, TC +0.030, ET +0.004)
258 (100 %)	0.778 ± 0.003	0.782 ± 0.002	0.779 ± 0.003	−0.003

About half of the raw +0.044 at 10 % real is the longer schedule (the control gains +0.023 on its own); the other half, +0.021, is attributable to the synthetic pairs and exceeds the seed spread of both conditions. At 25 % the control gains nothing and synthetic pre-training keeps +0.014. At 100 % neither helps. The claim in finding 6 is therefore kept at the corrected size.

Limitations

- 2-D slices only; no inter-slice consistency, and Dice is aggregated per patient over 2-D slices rather than computed on volumes.
- Inception features (natural images) for FID/KID; the real val-vs-test floor is reported to calibrate them, but a radiology-specific extractor would be more sensitive to clinically relevant detail.
- The VAE is frozen (`sd-vae-ft-mse`, trained on natural images); its ceiling bounds every model.
- One dataset (BraTS 2020, 369 patients, one split); the segmentation conclusions have 3 seeds but no external test cohort.
- Compute per model: 1.5 h (128 px) / 2.8 h (256 px) on one RTX A6000; 200 epochs each.

Appendix: auto-generated result tables

Result tables (auto-generated by scripts/collect_results.py)

Generative quality (samples vs real test slices)

run	samples	FID rgb ↓	KID rgb ×10³ ↓	FID flair	FID t1ce	FID t2	ref FID rgb (val vs test)	pair-SSIM (samples / real) ↓	NN dist (samples / real test)	memorised ↓
ldm128_maskcond	samples_best_ddim50_cfg1_seed0	26.33	21.27	47.87	45.14	70.51	10.49	0.582 / 0.652	9.33 / 9.96	0.068
ldm128_maskcond	samples_best_ddim50_cfg2_seed0	26.22	20.89	45.85	45.42	73.03	10.49	0.576 / 0.652	9.68 / 9.96	0.069
ldm128_maskcond	samples_best_ddim50_cfg2_seed0_valmasks	27.94	21.68	47.02	45.44	75.85	10.49	0.582 / 0.652	9.71 / 9.96	0.078
ldm128_maskcond_do01	samples_best_ddim50_cfg1_seed0	24.05	19.14	49.56	46.09	73.08	10.49	0.590 / 0.652	9.05 / 9.96	0.075
ldm128_maskcond_do01	samples_best_ddim50_cfg2_seed0	24.34	19.13	47.65	46.20	75.71	10.49	0.585 / 0.652	9.40 / 9.96	0.070
ldm128_maskcond_do01	samples_best_ddim50_cfg2_seed0_valmasks	25.55	19.34	48.79	45.48	77.51	10.49	0.586 / 0.652	9.43 / 9.96	0.087
ldm128_maskcond_reg	samples_best_ddim50_cfg1_seed0	20.37	14.74	46.95	44.23	69.37	10.49	0.586 / 0.652	10.60 / 9.96	0.036
ldm128_maskcond_reg	samples_best_ddim50_cfg1_seed0_ftdec	18.06	14.56	26.81	25.73	17.50	10.49	0.638 / 0.652	10.46 / 9.96	0.040
ldm128_maskcond_reg	samples_best_ddim50_cfg2_seed0	20.85	15.19	44.33	43.97	69.57	10.49	0.584 / 0.652	10.71 / 9.96	0.043
ldm128_maskcond_reg	samples_best_ddim50_cfg2_seed0_ftdec	20.89	18.91	28.61	25.62	17.57	10.49	0.636 / 0.652	10.49 / 9.96	0.051
ldm128_maskcond_reg	samples_best_ddim50_cfg2_seed0_valmasks	22.85	16.52	45.20	44.06	71.12	10.49	0.593 / 0.652	11.09 / 9.96	0.040
ldm128_maskcond_reg	samples_best_ddim50_cfg2_seed0_valmasks_ftdec	22.49	20.21	29.34	26.66	17.91	10.49	0.640 / 0.652	10.90 / 9.96	0.046
ldm128_uncond_reg	samples_best_ddim50_cfg1_seed0	21.31	15.87	47.54	45.00	68.68	10.49	0.579 / 0.652	11.20 / 9.96	0.027
ldm256_maskcond_reg	samples_best_ddim50_cfg1_seed0	12.83	8.77	43.14	41.72	58.23	9.55	0.637 / 0.690	10.93 / 9.90	0.021
ldm256_maskcond_reg	samples_best_ddim50_cfg2_seed0	10.45	6.40	40.29	39.64	59.07	9.55	0.628 / 0.690	11.03 / 9.90	0.030
ldm256_maskcond_reg	samples_best_ddim50_cfg2_seed0_valmasks	10.83	6.17	40.38	39.25	60.65	9.55	0.633 / 0.690	11.39 / 9.90	0.024

VAE reconstruction ceiling (real test slices; depends only on the data and the decoder, not on the diffusion model)

data	decoder	channel	PSNR (dB)	SSIM	LPIPS
brats128 (ldm128_maskcond)	frozen	flair	27.00 ± 1.93	0.839 ± 0.049	0.062
brats128 (ldm128_maskcond)	frozen	t1ce	27.17 ± 1.96	0.842 ± 0.046	0.072
brats128 (ldm128_maskcond)	frozen	t2	25.39 ± 2.28	0.817 ± 0.049	0.078
brats128 (ldm128_maskcond)	frozen	rgb	26.37 ± 1.92	0.833 ± 0.046	0.041
brats128 (ldm128_maskcond_reg)	fine-tuned (`vae_dec_brats128`)	flair	28.56 ± 2.02	0.888 ± 0.033	0.057
brats128 (ldm128_maskcond_reg)	fine-tuned (`vae_dec_brats128`)	t1ce	28.35 ± 1.94	0.882 ± 0.032	0.063
brats128 (ldm128_maskcond_reg)	fine-tuned (`vae_dec_brats128`)	t2	27.00 ± 2.33	0.880 ± 0.030	0.065
brats128 (ldm128_maskcond_reg)	fine-tuned (`vae_dec_brats128`)	rgb	27.79 ± 1.85	0.883 ± 0.030	0.032
brats256 (ldm256_maskcond_reg)	frozen	flair	29.65 ± 1.84	0.882 ± 0.035	0.066
brats256 (ldm256_maskcond_reg)	frozen	t1ce	29.76 ± 1.85	0.885 ± 0.032	0.064
brats256 (ldm256_maskcond_reg)	frozen	t2	27.95 ± 2.06	0.863 ± 0.035	0.077
brats256 (ldm256_maskcond_reg)	frozen	rgb	28.98 ± 1.80	0.877 ± 0.032	0.036

VAE decoder fine-tuning (test-slice reconstruction ceiling before / after; encoder unchanged)

run	data	epochs (best)	trainable params	PSNR before → after	SSIM before → after	LPIPS before → after
vae_dec_brats128	brats128	8 (8)	49.5M	26.37 → 27.79	0.833 → 0.883	0.041 → 0.032

Checkpoint curve: best-validation-loss vs last checkpoint (2,000 samples each; FID/KID vs real validation slices; memorised = fraction of samples closer to a training slice than 95 % of real held-out slices)

run	best epoch	val loss	FID val ↓	KID val ×10³ ↓	memorised ↓	last epoch	FID val	memorised
ldm128_maskcond	34	0.0754	29.89	18.18	0.086	200	21.81	0.966
ldm128_maskcond_do01	41	0.0747	28.82	17.24	0.093	200	21.99	0.891
ldm128_maskcond_reg	101	0.0710	24.65	12.86	0.049	200	24.43	0.115
ldm128_uncond_reg	116	0.0723	24.81	14.75	0.040	200	23.20	0.051
ldm256_maskcond_reg	71	0.0663	15.10	6.41	0.033	200	12.94	0.291

Model selection (validation data only): ldm128_maskcond_reg

Rule: lowest FID vs validation slices at the best-val-loss checkpoint among runs with frac_memorised <= 0.15.

candidate	best epoch	FID val ↓	KID val ×10³ ↓	memorised ↓
ldm128_maskcond	34	29.89	18.18	0.086
ldm128_maskcond_do01	41	28.82	17.24	0.093
ldm128_maskcond_reg **(chosen)**	101	24.65	12.86	0.049

Downstream segmentation `seg` (segmenter config `ldm128_maskcond.yaml`, synthetic pool from `ldm128_maskcond_reg`; per-patient Dice on held-out test patients, mean ± std over seeds; rows after the primary protocol are the secondary analyses of EXPERIMENTS.md)

training data	real patients	real slices	synthetic slices	seeds	WT	TC	ET	mean
10% real	26	1483–1705	0	3	0.801 ± 0.011	0.580 ± 0.023	0.509 ± 0.022	0.630 ± 0.016
10% real + synthetic	26	1483–1705	1845–2208	3	0.809 ± 0.006	0.559 ± 0.010	0.430 ± 0.033	0.600 ± 0.014
10% real (VAE-reconstructed)	26	1483–1705	0	3	0.786 ± 0.016	0.535 ± 0.029	0.397 ± 0.028	0.573 ± 0.023
10% real, real pre-training (control)	26	1483–1705	0	3	0.805 ± 0.010	0.604 ± 0.017	0.549 ± 0.018	0.653 ± 0.010
10% real, synthetic pre-training	26	1483–1705	0	3	0.830 ± 0.003	0.628 ± 0.013	0.564 ± 0.012	0.674 ± 0.008
10% real + synthetic 1:1	26	1483–1705	1483–1705	3	0.816 ± 0.002	0.572 ± 0.008	0.442 ± 0.007	0.610 ± 0.006
25% real	64	3937–4031	0	3	0.847 ± 0.003	0.676 ± 0.011	0.611 ± 0.010	0.711 ± 0.004
25% real + synthetic	64	3937–4031	4913–5158	3	0.844 ± 0.004	0.660 ± 0.021	0.521 ± 0.014	0.675 ± 0.012
25% real (VAE-reconstructed)	64	3937–4031	0	3	0.822 ± 0.005	0.621 ± 0.009	0.475 ± 0.006	0.639 ± 0.003
25% real, real pre-training (control)	64	3937–4031	0	3	0.849 ± 0.001	0.670 ± 0.012	0.616 ± 0.009	0.712 ± 0.003
25% real, synthetic pre-training	64	3937–4031	0	3	0.858 ± 0.003	0.700 ± 0.010	0.620 ± 0.006	0.726 ± 0.005
25% real + synthetic 1:1	64	3937–4031	3937–4031	3	0.844 ± 0.007	0.668 ± 0.018	0.549 ± 0.008	0.687 ± 0.011
100% real	258	15895	0	3	0.886 ± 0.001	0.775 ± 0.005	0.673 ± 0.004	0.778 ± 0.003
100% real + synthetic	258	15895	20000	3	0.879 ± 0.001	0.742 ± 0.009	0.650 ± 0.010	0.757 ± 0.006
100% real (VAE-reconstructed)	258	15895	0	3	0.862 ± 0.006	0.713 ± 0.008	0.542 ± 0.026	0.706 ± 0.006
100% real, real pre-training (control)	258	15895	0	3	0.888 ± 0.003	0.779 ± 0.002	0.678 ± 0.006	0.782 ± 0.002
100% real, synthetic pre-training	258	15895	0	3	0.889 ± 0.002	0.770 ± 0.007	0.677 ± 0.001	0.779 ± 0.003
100% real + synthetic 1:1	258	15895	15895	3	0.878 ± 0.004	0.757 ± 0.007	0.643 ± 0.013	0.760 ± 0.004
synthetic only	0	0	20000	3	0.829 ± 0.002	0.669 ± 0.007	0.481 ± 0.012	0.660 ± 0.005

Mask consistency of synthetic samples (per-slice Dice of a real-trained segmenter vs the conditioning mask, mean ± std over seeds)

segmenter	n synthetic	WT	TC	ET
10% real	20000	0.767 ± 0.006	0.506 ± 0.023	0.439 ± 0.025
25% real	20000	0.783 ± 0.003	0.566 ± 0.006	0.493 ± 0.012
100% real	20000	0.789 ± 0.002	0.584 ± 0.006	0.494 ± 0.004

Downstream segmentation `seg256` (segmenter config `ldm256_maskcond_reg.yaml`, synthetic pool from `ldm256_maskcond_reg`; per-patient Dice on held-out test patients, mean ± std over seeds; rows after the primary protocol are the secondary analyses of EXPERIMENTS.md)

training data	real patients	real slices	synthetic slices	seeds	WT	TC	ET	mean
10% real	26	1483–1705	0	3	0.849 ± 0.006	0.672 ± 0.019	0.604 ± 0.009	0.709 ± 0.003
10% real + synthetic	26	1483–1705	1845–2208	3	0.860 ± 0.006	0.721 ± 0.007	0.626 ± 0.006	0.736 ± 0.004
25% real	64	3937–4031	0	3	0.874 ± 0.003	0.751 ± 0.009	0.668 ± 0.010	0.764 ± 0.007
25% real + synthetic	64	3937–4031	4913–5158	3	0.878 ± 0.000	0.769 ± 0.004	0.654 ± 0.000	0.767 ± 0.001
100% real	258	15895	0	3	0.895 ± 0.000	0.810 ± 0.002	0.707 ± 0.002	0.804 ± 0.000
100% real + synthetic	258	15895	20000	3	0.894 ± 0.004	0.818 ± 0.006	0.692 ± 0.008	0.801 ± 0.005
synthetic only	0	0	20000	3	0.871 ± 0.002	0.785 ± 0.008	0.623 ± 0.004	0.760 ± 0.003

Mask consistency of synthetic samples (per-slice Dice of a real-trained segmenter vs the conditioning mask, mean ± std over seeds)

segmenter	n synthetic	WT	TC	ET
10% real	20000	0.848 ± 0.003	0.616 ± 0.025	0.567 ± 0.042
25% real	20000	0.864 ± 0.001	0.699 ± 0.014	0.665 ± 0.029
100% real	20000	0.875 ± 0.000	0.756 ± 0.011	0.728 ± 0.009

Downstream segmentation `seg_ftdec` (segmenter config `ldm128_maskcond_reg.yaml`, synthetic pool from `ldm128_maskcond_reg`; per-patient Dice on held-out test patients, mean ± std over seeds; rows after the primary protocol are the secondary analyses of EXPERIMENTS.md)

training data	real patients	real slices	synthetic slices	seeds	WT	TC	ET	mean
10% real	26	1483–1705	0	3	0.796 ± 0.009	0.579 ± 0.026	0.510 ± 0.025	0.628 ± 0.017
10% real + synthetic	26	1483–1705	1845–2208	3	0.802 ± 0.016	0.578 ± 0.015	0.446 ± 0.019	0.609 ± 0.006
25% real	64	3937–4031	0	3	0.848 ± 0.002	0.683 ± 0.006	0.612 ± 0.013	0.714 ± 0.005
25% real + synthetic	64	3937–4031	4913–5158	3	0.845 ± 0.003	0.668 ± 0.008	0.530 ± 0.011	0.681 ± 0.007
100% real	258	15895	0	3	0.885 ± 0.002	0.770 ± 0.001	0.674 ± 0.006	0.776 ± 0.002
100% real + synthetic	258	15895	20000	3	0.883 ± 0.001	0.759 ± 0.007	0.652 ± 0.004	0.765 ± 0.004
synthetic only	0	0	20000	3	0.832 ± 0.006	0.690 ± 0.004	0.495 ± 0.018	0.672 ± 0.004

Mask consistency of synthetic samples (per-slice Dice of a real-trained segmenter vs the conditioning mask, mean ± std over seeds)

segmenter	n synthetic	WT	TC	ET
10% real	20000	0.773 ± 0.005	0.527 ± 0.027	0.491 ± 0.039
25% real	20000	0.791 ± 0.002	0.597 ± 0.013	0.555 ± 0.014
100% real	20000	0.796 ± 0.001	0.648 ± 0.010	0.586 ± 0.007